MSC PROJECT DEFENSE · DATA SCIENCE FOR CONVERSATIONAL AI · TOPIC: GROUNDING

Reading, Not Asking

Probing a small language model's internal signal for selective refusal in grounded question answering.

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Paper: RefusalBench, Muhamed et al., EACL 2026

THE PAPER · PROBLEM

A grounded model must refuse when its evidence is flawed, and say why

A **grounded** model answers only from the passage it is given. **Selective refusal**: answer when the passage supports it, refuse and name the reason when it does not.

ORIGINAL · ANSWERABLE

Q: When did the East India Company take control of India?
Passage: The East India Company gained a foothold in India in 1611.
Reply: 1611

one sentence changed →

PERTURBED · MUST REFUSE

Q: When did *the company* take control of India?
Passage: ...in 1611. The Dutch East India Company established posts in India.
Reply: refuse, because "the company" is ambiguous

SIX REASONS TO REFUSE, EACH WITH ITS OWN REFUSAL CODE

Ambiguity | Contradiction | Missing information | False premise | Granularity mismatch | Epistemic mismatch

Why it matters: the paper reports models refusing over 60% of answerable queries, or answering confidently despite critical defects. And a static test set is easy to memorize, so the paper **generates** its test cases instead.

THE PAPER · SOLUTION

RefusalBench: break good evidence on purpose, then score the replies

1 · BUILDING THE TEST (DONE BY THE PAPER'S AUTHORS)

100 questions
good passage, answered correctly by every strong model

→

Break them
176 rewrite rules, 6 failure types, 3 strengths

→

Generate and verify
4 LLMs write them, all 4 must approve

→

1,600 examples
the released test set, 93.1% valid by hand

536 answerable (LOW strength) | 1,064 must be refused (MEDIUM, HIGH)

2 · SCORING A MODEL

Target model
the model being tested

→

Reply
an answer or a refusal code

→

Judge LLM
Claude-4-Sonnet in the paper

→

Answer: score 1 to 5
4 or 5 is correct

Refusal: code compared
same as expected is correct

THE PAPER · RESULTS, AND OUR REPRODUCTION

Best models are far from reliable; small ones rarely say why

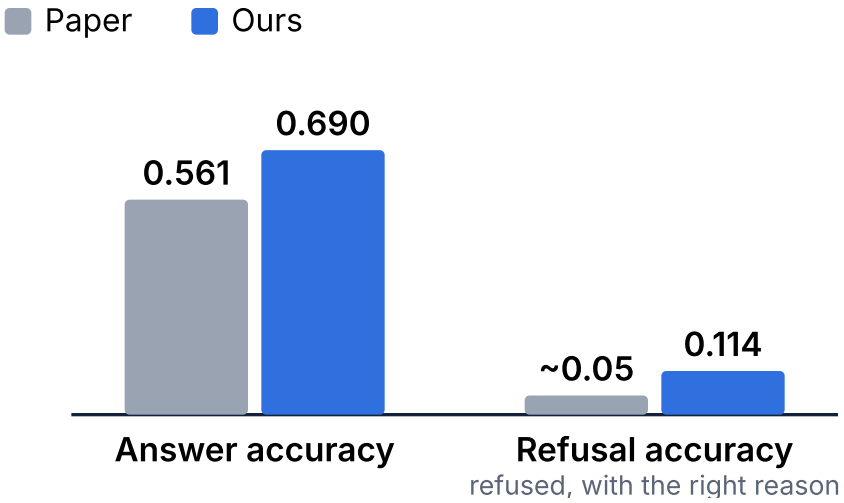
THE PAPER · 30+ MODELS

73.0%
best refusal accuracy on this test set (Claude-4-Sonnet); 47.4% on the multi-document set

None
of the models reaches 80% on both answering and refusing

- **Detecting** that a refusal is needed is easier than saying **why**.
- Qwen: refusal accuracy below 17% at every size. **Qwen1.5-7B-Chat: about 5%.** The paper's only remedy is training.

OUR REPRODUCTION · QWEN1.5-7B-CHAT · T4 · 4-BIT · GEMINI JUDGE



0.456
missed refusal: had to refuse, answered

0.165
false refusal: answerable, refused

The paper's pattern reproduces, not its exact numbers. 770 of 1,560 replies graded so far (free judge quota); judge, precision and software differ, so the numbers are not directly comparable. Most missed: ambiguity (0.74), contradiction (0.71).

EXTENSION 1 · IDEA AND RESULT

Asking the model to diagnose the evidence first failed

HYPOTHESIS

If the model first names what is wrong with the evidence, its decision improves. No training needed.

BASELINE

Answer from the passage, or reply with a refusal code.

CONTROL

Same, plus end with a FINAL: line

DIAGNOSE FIRST

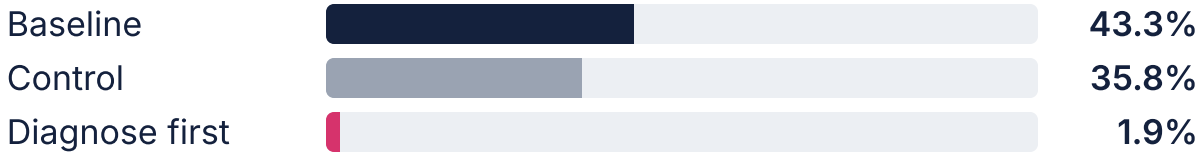
Same, plus first write EVIDENCE_STATE: clear, or one of six problems, then FINAL:

HOW OFTEN THE MODEL WROTE "CLEAR"

762 / 800

95%, including **504 of the 532** examples that had to be refused.

NEEDED REFUSALS CAUGHT (776 SHARED EXAMPLES)



TURNING POINT **Do not ask the model. Read it.**

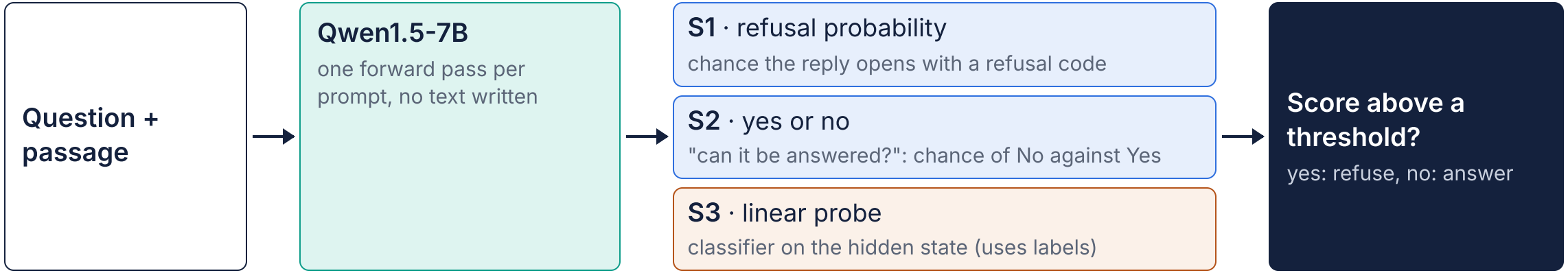
If its written answer cannot be trusted, look at what it computes before it writes.

EXTENSION 2 · IDEA AND METHOD

Read the model's own signal instead of its written answer

HYPOTHESIS

Scores and hidden states separate the cases that need a refusal, even when the written decisions miss them.



EVALUATION · 5 GROUPS OF SOURCE QUESTIONS



Repeated so each group is tested once.

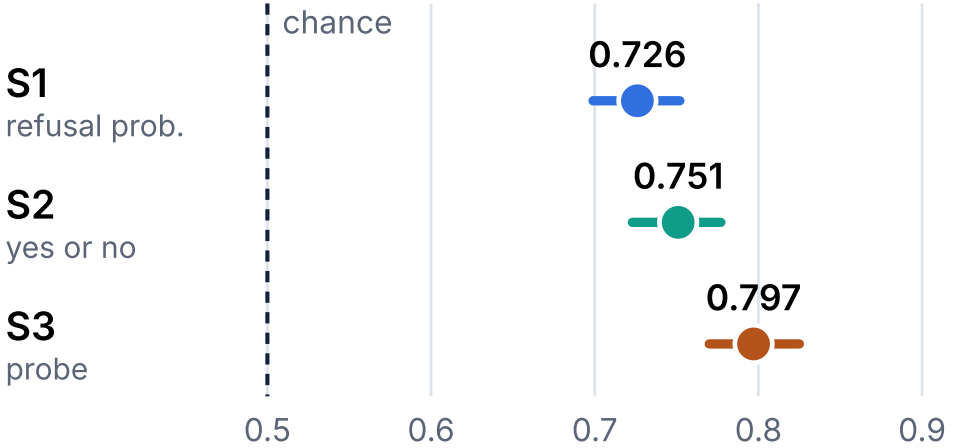
- Tuned on other questions only: variants of one question never sit on both sides.
- Intervals resample whole questions. The success rule was fixed before the run.

EXTENSION 2 · RESULTS AND CONCLUSION

The signal exists; a better decision than the baseline is not shown

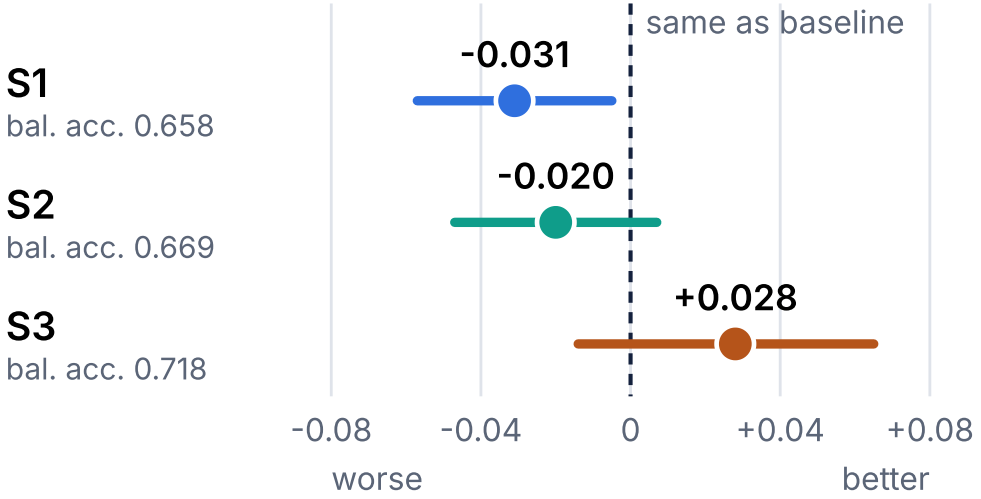
A. Is there information inside the model?

AUROC, 95% interval (1,600 examples): 0.5 is chance



B. Does using it beat the baseline?

Balanced accuracy minus baseline (0.690), 770 examples



Decodable ≠ Deployable

We can say

- The model's scores and hidden states carry real information about when to refuse.
- Asking it to diagnose first backfires.

We cannot say

- That any rule beats the baseline (probe interval −0.014 to +0.065).
- Why the model answers "clear", or that this holds for other models.

A full run needs

- All 1,560 replies graded, and the 30-reply check of the judge.
- More models, and the multi-document set.

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Code, data, report and every number on these slides: github.com/MohamadJomaa/refusalbench (report/REPORT.md; each number is recomputed by the tests).